FINAL REPORT – CUSTOMER CHURN PREDICTION PROJECT

E-Commerce Company | Capstone Project | Final Submission | Great Learning

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Executive Summary

This report presents the most important findings of the customer churn prediction project developed for an e-commerce company that is facing an increasing number of client churn. The company wanted a churning prediction model to increase early intervention, protect revenue, and provide ways to increase operational efficiency. With the data provided, we were able to develop and analyze four models, with the goal of selecting the one that offered the best results.

The first step in building the model was to understand the data (its structure and characteristics) and define the business problem. The data was analyzed using exploratory data analysis and data preprocessing techniques. The results and findings of both of those processes are detailed in the EDA and Business Implications and Data Cleaning and pre-processing sections of the report.

The second step was the creation and definition of the four models. The models were compared and analyzed to find the one that best fit the company's needs and the project's main goal.

Vocabulary

CONCEPT	DEFINITION
Churn	For this document, churn is defined as the proportion of customers who stop their relation with the business.
Exploratory Data Analysis (EDA)	The process of analyzing and visualizing a set of data to understand its structure and identify patterns and insights from it.
Chi-Square Test	The method is used to check whether the differences between groups are meaningful.
Mann-Whitney U Test	A test that checks whether two groups show different numerical behaviors.

Table 1Vocabulary

Introduction

The client has been having problems with customer retention: the data shows a rising customer churn rate (see Chart 1). The identified causes of the attrition in the customer base are intense competitive pressure in the industry, increasing customer expectations, and a negative service rating that reflects high service dissatisfaction. Figure 1 shows the main factors contributing to churn and illustrates why a predictive approach is essential for the company. As of today, the company lacks a churn detection system to create actions to prevent attrition before disengagement from the customer. Chart 1 shows the level of churn that the company has.

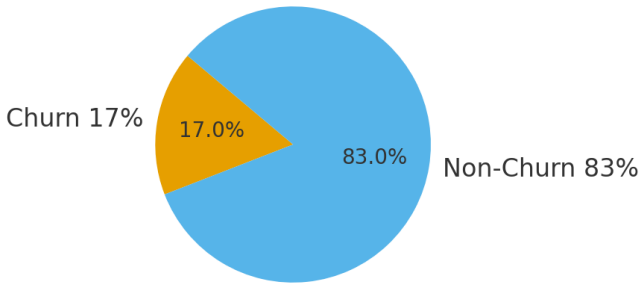


Chart 1Churn percentage



Figure 1Churn dynamics

Scope: Analyze customer behavior and service interaction data to identify the key drivers of churn and develop a predictive model.

Problem Definition: Develop an account churn prediction model based on computer data to estimate churn risk. The system should provide enough information for the company to develop proactive retention strategies and support decision-making

Objective: Detect churn early to support earlier actions to prevent disengagement.

Figure 2 summarizes the analytical workflow followed for the creation of the model and outlines the structure of the report.

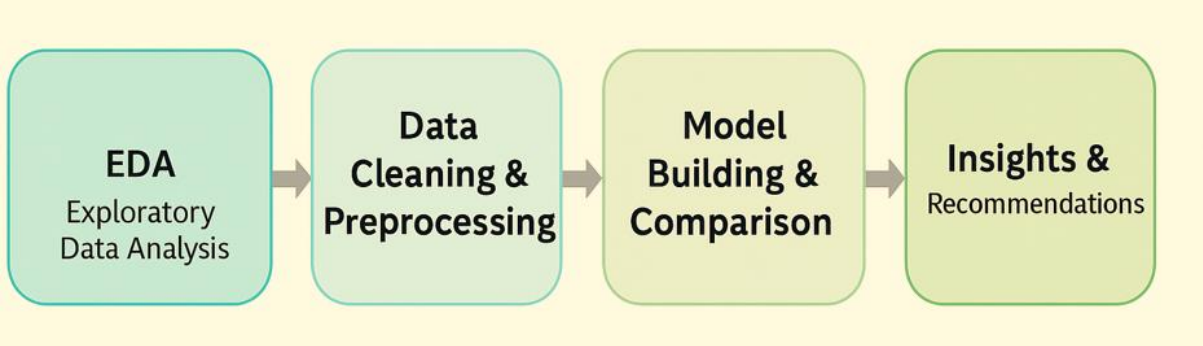


Figure 2 Analytical Blueprint

Exploratory Data Analysis and Business Implications

The EDA analysis help understand the patterns in customer behavior before we begin to build the predictive model. The analysis shows how churn relates to other variables. was conducted at three levels -univariate, bivariate, and multivariate- to analyze the data and explore the findings, behaviors and meaningful patterns that can be translated into business insights. The following are the key findings of each analysis.

Univariate Analysis Findings

A univariate analysis examines each variable individually and tries to understand what it says about the customer's behavior. Table 2 presents the key findings of the univariate analysis.

Category	Conclusions	Business Meaning
Customer Tenure	Most customers are early in their lifecycle.	New customers are significantly more vulnerable to churn.

Customer Value & Spending	Many customers spend little and show limited growth.	Small declines in a large base can result in meaningful revenue loss.
Service Experience	Service and support scores are mostly average.	“Average service” isn't enough to prevent churn — quality issues cause early exits.
Customer Complaints	A small group has filed complaints.	Even a single complaint significantly increases churn risk; complaints must trigger immediate intervention.
Engagement & Activity	Many customers have little or no recent activity/contact.	Early disengagement is a silent churn driver; inactivity signals risk.
Account Characteristics	Many accounts have multiple users; a minority are single user.	Single-user accounts require targeted retention; multi-user accounts are more stable.
Demographics & Preferences	Most customers are Tier 1, mobile-first, debit-card users.	Retention strategy must be optimized for mobile-first Tier 1 users.

Table 2 Univariate Analysis summary

Bivariate Analysis

A bivariate analysis compares two variables at the same time to understand how they are related: it essentially looks at two variables and sees if changes in one are linked to changes in

the other. Table 3 shows the key findings of the bivariate analysis for the customer categories variables.

Variable Group	What We Observed (Simple)	Business Implication
CUSTOMER CATEGORIES (Comparing group differences)		
Tenure Group	Newer customers leave more often.	Strengthening onboarding and early engagement.
Customer Complaints	One complaint strongly increases churn risk.	Complaints must trigger immediate intervention.
Service Experience (Service Score, Agent Score)	Lower service scores appear more in churners.	Improving service quality reduces churn.
Account Type (Single vs Multi-User)	Single-user accounts churn more than multi-user.	Prioritize retention for single-user accounts.
Customer Segment	Some segments show higher churn rates.	Segment-based retention improves effectiveness.

Payment Method	Some payment types show higher churn.	Payment behavior can signal loyalty patterns.
Login Device	Mobile-only customers show different churn tendencies.	Mobile experience is critical for retention.
City Tier	Churn varies across city tiers.	Enables geographic prioritization of campaigns.

Table 3 Bivariate Analysis customer categories variables

When comparing customer groups, some clear patterns emerged linking certain behaviors to churn (our key variable in the project). All the major customer categories show important differences between customers who stay with the company and those who leave. What this means is that churn is not a random behavior; it changes across customer groups depending on their behaviors. This analysis was made using the chi-square test, which is a method to determine if different customer groups behave differently.

When the numerical factors (those that refer to behaviors or activities that can be measured numerically) were analyzed, they had the same type of difference: customers who left showed distinct patterns compared with those who stayed. The numerical factors were analyzed based on the results of the Mann-Whitney U test, which compares two sets of customers based on numerical measures to show if one group consistently behaves differently.

Variable Group	What We Observed (Simple)	Business Implication
NUMERICAL FACTORS (Comparing numeric differences)		
Tenure (Months)	Churners have fewer months with the company.	New customers need targeted support.
Customer Care Interactions	Churners show different patterns of support usage.	Support behavior is an early indicator of churn.
Complaint Count	Churners have more recorded complaints.	Complaint prevention is key to retention.
Monthly Revenue	Churners often spend less.	Low spend may signal declining interest.
Revenue Growth YoY	Churners show weaker or negative growth.	Revenue decline is an early warning of churn.

Table 4 Bivariate Analysis numerical factors

Multivariate Analysis

A multivariate analysis investigates how different customer variables relate to each other and work together. This analysis indicated that there was no single numeric measure that could explain churn behavior, which means that it is driven by a complex combination of behaviors. The analysis also revealed that the combination of some customer groups' characteristics leads to different behaviors.

The conclusion is that the numerical correlation analysis showed there were no strong linear correlations, and that is influenced by a combined number of variables like demographics, service preferences, and engagement patterns.

Business Implications

To translate the EDA findings into business insights, a matrix that combines business impact and evidence strength was created. The business impacts reflect how strongly the variables influence churn behaviors, and the EDA strength reflects how strong the evidence was in the analysis. The results of the matrix are shown in Figure 1 and the insights in Table 5. From a business perspective, the conclusion is that churn is a multi-factor phenomenon, and behavior, satisfaction, experience, and demographic segments all play an important role. Customers who feel unimportant or unsupported, as well as new customers lacking established habits and customers with poor or declining usage, are risk profiles.



Figure 3 Business Impact Evidence Strength Matrix

Quadrant	Business Interpretation
High Business Impact + Strong Evidence (Top-Right Quadrant)	These variables represent the most influential churn drivers and show clear, consistent separation between churners and non-churners in the EDA. They are the primary levers where management action will produce the fastest and most meaningful reduction in churn.
Medium Business Impact + Strong Evidence	These variables show strong visual and statistical evidence, but their business impact is more supportive than primary. They function as early-warning signals that help identify customers who are beginning to disengage or experience service issues. They should be integrated into monitoring and retention workflows.
Medium Business Impact + Medium Evidence	It shows moderate differences between churn groups. Although not a direct churn driver, it is essential for prioritizing high-value customers at risk, ensuring that retention resources are allocated effectively.
Low Business Impact Quadrants	These variables show measurable but lower influence on churn. They are best used for segmentation, tailoring messaging, or designing channel-specific retention incentives rather than guiding core churn decisions.

Table 5 Impact and Evidence Strength Matrix Interpretation

Data Cleaning and Pre-processing

Data cleaning and preprocessing are a group of actions taken to clean the data so that all the models work on the most accurate information. These actions help eliminate all the inconsistencies, determine and treat incomplete records accordingly, and prepare the information

so that it reflects the actual behavior of the customer and guarantee that the churn analysis will be built on consistent and clean data. Figure 2 shows the process used to perform these actions.

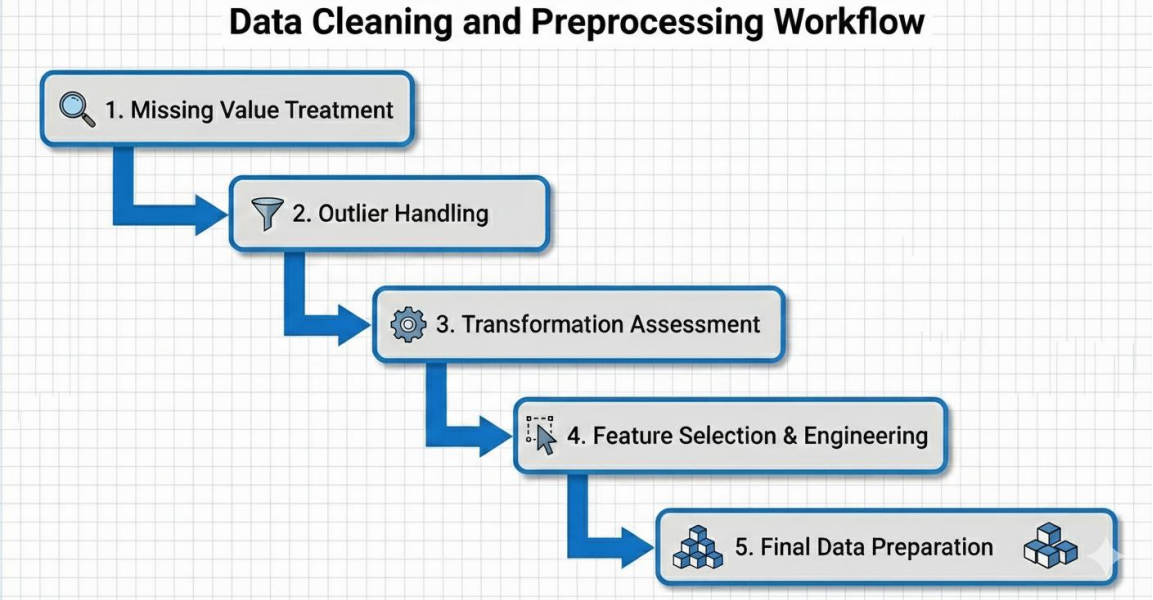


Figure 4 Data Cleaning and Preprocessing Workflow

Missing Value Treatment:

Most likely due to onboarding gaps or uncaptured service information, some customer records contained incomplete data. Instead of using a single value for the entire dataset, the missing values were handled in such a way that they were completed based on the behavior of similar customers (this is called conditional imputation). The numerical fields were filled using the median within the appropriate customer group and the categorical fields were completed using the most frequent category within the same customer group. The approach used makes the data set more reliable because it respects how similar customers behave.

Outlier treatment:

Outliers are customer records that are very different from most of the data. In this project, an outlier is a customer whose behavior is atypical and is relevant because they often represent

the most valuable or highest-risk customers. By keeping them, the analysis will reflect the full range of customer behavior.

Transformation assessment

Before building a model, the data was reviewed to determine whether any numerical variable needed to be mathematically adjusted, which happens when the data is very uneven (for example, if many customers fall into one range and only a few in very high or very low ranges).

In this project, that transformation was not significant because the natural customer patterns were meaningful, and the techniques used to build the models handled uneven data very well.

Feature Selection and Engineering

The next step is to decide which customer information most clearly explains churn behavior and which fields only add noise.

First, we chose what we want to keep or remove. This decision is important because not all customer fields are useful. Then we did feature engineering, which is the process of enhancing the meaningful variables or keeping them as they were.

Identifier fields, variables with almost no variation, and fields containing post-churn information were removed from the dataset because they do not add any value, contribute in a meaningful way, or would introduce bias. Behavioral, financial, and experience-related variables, like customer complaints, revenue growth, inactivity, and service scores, were not removed and were kept because the EDA analysis showed they can strongly differentiate churners from non-churners.

Final Data Preparation:

Once all the data was reviewed, the final part was to take all the cleaned fields and assemble them into a new dataset for analysis and modeling. This dataset represents a realistic picture of the customer base and its behaviors, and it is ready to build a model that can be trusted to make informed decisions.

Model Building and Interpretation

The modeling building focused on trying to understand which analytical approaches were the most effective at identifying customers at risk of churn.

Model Building Approach

A multiple model analysis was used, and four prediction methods were tested and developed to compare how well each one identifies customers at risk. Because different models learn different models, this approach helps to understand how each one interprets and captures customer behavior and how consistent they are in identifying churn. This approach ensures reliability and strengthens business confidence in the analytical process.

Using models with different logics allowed us to see how consistently the data set indicates credit risk and if some customer characteristics are reliable across all the analytical approaches. The models are shown in Table 6.

Model	Business Description	Selection Criteria
KNN	Simple pattern-matching model that compares customers to similar past cases.	Helps answer the question Is churn predictable just by analyzing customers who have similar patterns?
Decision Tree	Learns easy-to-understand decision rules based on customer behavior.	Mimics human decision-making by splitting customers based on specific conditions using sequential questions. Explains churn using simple and specific rules
Random Forest	Combines many decision trees to identify broader behavior patterns and reduce the risk of relying on a single decision path.	Validates whether churn signals are stable across many scenarios rather than relying on a single rule set.
XGBoost	Advanced model that captures complex patterns and interactions for the strongest early warnings.	Chosen because it improves itself iteratively, focusing on the customers who are hardest to classify, making it extremely strong at detecting subtle churn drivers.

Table 6 Description of Models

Evaluation Metrics Rationale

To compare the models and evaluate how well each model identifies customers at risk of leaving, three performance metrics were reviewed: recall, precision, and accuracy. Each of these metrics captures a different aspect of model performance and helps explain if a model

is better and useful for the business need at hand. Table 7 offers a description of each metric and the business problem interpretation.

Metric	Description	Business Iterpretation
Recall	% of customers who actually churned that the model correctly identified.	Of all the customers who truly left, what percentage did we successfully flag as at risk? This reflects the model's ability to prevent customers from slipping away unnoticed.
Precision	% of customers flagged as churners that were actually churners.	Of the customers we decide to focus retention actions on, how many really were at risk? This reflects the efficiency and cost-effectiveness of interventions.
Accuracy	Overall correctness, or % of all predictions that the model got right.	How often is the model right overall? But in churn (an imbalanced dataset), accuracy can look high even when the model misses many churners — so it is not used as the main evaluation metric.

Table 7 Metrics Description and Interpretation

As it was stated earlier, churn is an imbalanced problem where the majority of the customers stay and only a smaller portion leaves. From a business standpoint, the churn rate is high, and the main objective is to reduce it by identifying clients early. This issue is

important because some metrics can appear strong but are really failing to detect churn earlier.



Figure 5Metric description

When evaluating a model, two questions have to be answered: how many real customers at risk the model identifies successfully, and how often the alerts it generates are genuinely actionable. So the selected model must capture as many true churners as possible while the retention actions remain focused and at an affordable cost. Table 7 compares the metrics and how each one aligns with the company's business objective of reducing preventable churn.

Metric	What It Measures	Why It Fails for Churn	Alignment With Business Objective	Impact on Churn Prevention
Accuracy	Overall correctness across all predictions.	Can look high even when the model fails to detect most churners.	Does not align with the business goal of identifying at-risk customers.	Low — hides the real churn problem and gives a false sense of confidence.
Precision	Of the customers flagged as churners, how many truly churned.	Does not measure how many actual churners were missed.	Helps with cost control, but not with the main goal of catching churners early.	Medium — efficient but not protective enough.
Recall (chosen)	How many true churners the model successfully identified.	None — recall directly focuses on detecting churners.	Fully aligned with the business objective of reducing preventable churn.	High — maximizes early detection of at-risk customers and protects revenue.

Table 8Metric Analysis

As shown in the table, while accuracy and precision offer useful information, they do not address the main financial risk and fail to identify customers who are contemplating leaving the company. On the contrary, recall is the only metric that directly measures how

many real churners a specific model identifies correctly. That is the main reason that recall was the metric prioritized to evaluate the models.

Model Improvement and Finalizing the Model

After the models were created, they were tested. Table 9 shows a summary of the initial performance of all the models.

Model	Recall(How many real churners we catch)	Precision(How many flagged churners are truly at risk)	Accuracy(Overall correctness)	Conclusion
KNN	0.766	0.884	0.944	❌ Too risky — misses many churners
Decision Tree	0.709	0.788	0.919	❌ Unstable — inconsistent with real customers
Random Forest	0.737	0.936	0.947	⚠️Better, but still not strong enough
XGBoost	0.912	0.881	0.978	⚠️Strong potential — needs tuning to improve recall further

Table 9 Model results before tuning

Performance Improvement

Based on the initial results, XGBoost was the model selected for tuning. KNN and Decision Tree were not selected because they performed significantly worse and showed limitations, indicating they were poor candidates for further improvement. And even though the Random Forest model showed strong potential, its recall numbers were moderate. Given the fact that XGBoost showed higher recall and stronger alignment with the business goal, all optimization resources were focused on the model most capable of delivering the highest value for the business.

To boost the prediction capabilities of the selected model, we used methods that combine the knowledge of many smaller models, instead of relying on one alone (this is called ensemble techniques).

Model	Recall (How many real churners we catch)	Precision (How many flagged customers are truly at risk)	Accuracy (Overall correctness)	Conclusion
KNN	0.766	0.884	0.944	❌ Too risky Misses many churners — high revenue leakage
Decision Tree	0.709	0.788	0.919	❌ Unstable Very unstable — fails with real customers
Random Forest	0.737	0.936	0.947	⚠️ Good but not best Reliable but not the safest for churn detection
XGBoost	0.874	0.953	0.972	✓ Best model Maximizes customer retention — highest business value

Image 1 Model Performance

Image 1 shows the result of the models after tuning. The results confirm that XGBoost was the best model because

- XGBoost detects the highest number of churn-risk customers.
- It generates minimal false alarms

- Recall is prioritized because missing churners is far more costly than contacting a few extra customers.

Table 9 shows a summary of the reasons for selecting this model, and Chart 1 shows the model recall comparison.

Reason	XGBoost Evidence	Business Importance
Highest Recall	0.874	Identifies most at-risk customers
Strong Generalization	Stable on new data	Reliable for real-world use
Handles Complex Patterns	Boosted trees	Captures subtle churn behavior
Efficient Learning	Low overfitting	Predictions are trustworthy

Table 10 XGBoost selection analysis

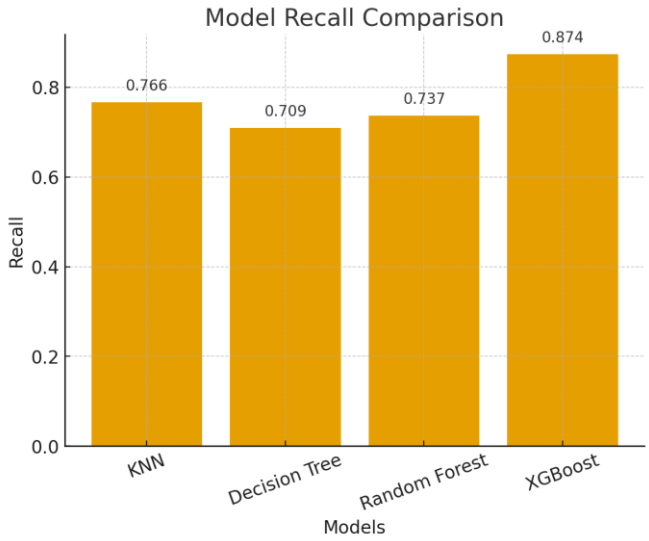


Chart 2Model Recall Comparison

Insights and Recommendations

Key Churn Drivers:

Insights derived from the model indicate that the primary drivers of churn include low tenure, recent complaints, low service scores, declining revenue patterns, and recent customer care interactions. Customers in certain segments and payment categories also show elevated risk.

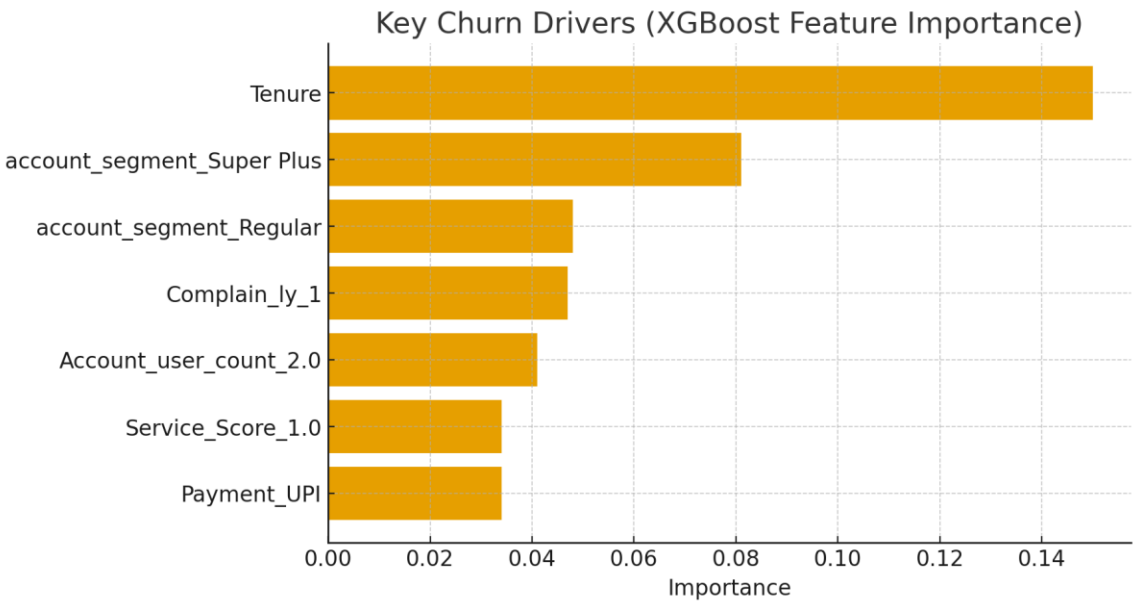


Chart 2 Key Churn Drivers

Tenure is the strongest churn driver

The data clearly shows that customers with shorter tenure are more likely to churn. This insight suggests that the company should focus on strengthening onboarding, early communication and proactive support in the first months of service to build engagement.

Multiple customer care interactions signal unresolved friction.

EDA indicates that those customers who contact support more than one time within a short period of time exhibit a strong likelihood of churning. This problem may require new protocols and a more effective workflow for unresolved problems.

Low service scores as a strong predictor of attrition:

Customers reporting low service or experience scores consistently churn more often than those with positive ratings, confirming that service dissatisfaction has direct financial implications. Management should monitor low-score segments closely and offer proactive support, service improvements, or tailored incentives to improve their experience.

Payment method influences churn likelihood:

Some payment methods, especially those requiring manual transactions, are associated with significantly higher churn rates, suggesting that billing friction plays a meaningful role in customer loss. The business should encourage customers to adopt more stable, automated payment methods while addressing renewal or card-update barriers.

Model confirmation that churn is multifactorial and predictable:

The XGBoost model identified Tenure, Complaints, Service Score, Inactivity, Customer Care Calls, and Revenue Growth as the strongest predictors of churn, perfectly aligning with EDA observations. This consistency confirms that churn follows predictable behavioral patterns and that these signals can be operationalized to guide targeted retention strategies.